Lesson plan · for the teacher

Reading data safely

Data science · Level 1 · Lesson 4

Time
40 min

Materials
A claim from a newspaper, advert or poster; Paper for group answers; A phone or tablet with Darsi, for the lab

Learning goals

- Look inside groups before comparing averages.
- Ask what else could explain a pattern.
- Share results without exposing a child.

Activity: What else could explain it?

1. Warm-up (5 min): Read a claim such as “Pupils who sit at the front get better marks.” Ask: should we move everyone to the front?
2. Teach (8 min): Tell the morning and afternoon story, then the breakfast example. Model the question “What else could explain it?”
3. Check (15 min): Each group gets a claim card and writes two other explanations and the data that would check the claim.
4. Privacy (7 min): Show four ways of sharing results (names on the wall, the class average, a file with numbers, a ranked list). Pupils sort them into safe and unsafe.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

Discussion

- What could explain the front-row pattern, apart from the seat?
- Why is a class of eight risky to judge on one test?
- How will you share the next test’s results?

Exit ticket

- Write one pattern from school life and one other thing that could explain it.

Try it in class tomorrow
Bring one claim from a newspaper or a poster. Ask your pupils what else could explain it, and what data would check it.

Worksheet · for pupils

Reading data safely

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Name _____ Class _____ Date _____

1 **What is the safest way to show parents how the class did?**

◯ Every pupil's mark, by name ◯ The class average and spread

◯ A ranked list from best to worst

2 **If the morning class has a higher average, the time of day must be the reason.**

◯ True ◯ False

3 **Each pair is (pupil number, mark). Print only the class mean, rounded to one decimal place, so the result can be shared without any child's mark.**

```
marks = [("P01", 14), ("P02", 9), ("P03", 17), ("P04", 12)]
for pupil, mark in marks:
    print(pupil, mark)
```

Write your program:

4 **Before acting on "the morning class does better", put these checks in a sensible order.**

☐ Only then decide what to change

☐ Look at more than one test

☐ Compare similar pupils across the classes

☐ Ask who is in each class

5 **Which habits keep pupils' data safe? Choose all that apply.**

☐ Use pupil numbers instead of names ☐ Collect only what the question needs

☐ Pin a ranked list on the wall ☐ Delete the data when you're done

✎ **Where does your school share pupils' results today? Is any child exposed?**